



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

COMPILER DESIGN-CSA1402

CO1_AT2 ERROR ANALYSIS

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Problem 1:

The following program is intended to perform simple arithmetic operations, function calls, array access, and looping constructs. However, the code contains multiple errors introduced at different phases of compilation, including lexical, syntax, semantic, logical, and runtime errors.

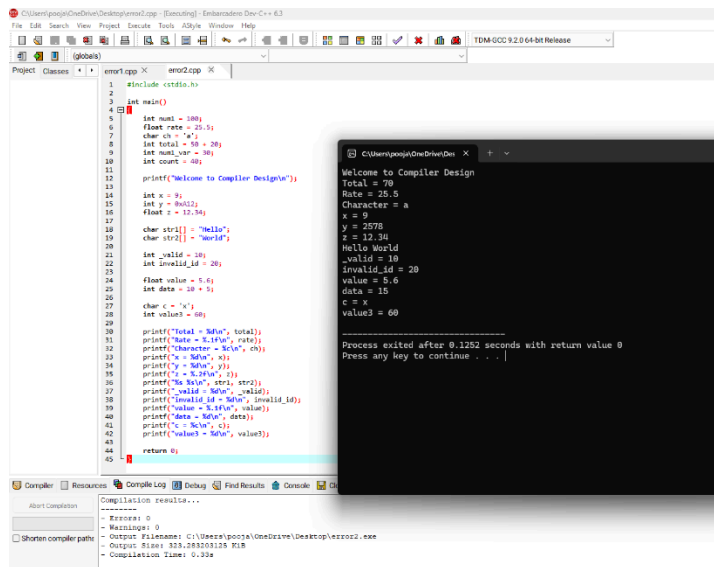
```
1 #include <stdio.h>
2
3 int sumarray(int arr[], int n)
4 {
5     int i, sum = 0;
6     for(i = 0; i < n; i++)
7     {
8         sum += arr[i];
9     }
10    return sum;
11 }
12
13
14
15 int main()
16 {
17     int arr[5] = {1,2,3,4,5};
18     int total;
19     total = sumarray(arr, 5);
20
21     if(total == 15)
22     {
23         printf("Correct Sum\n");
24     }
25
26     int a = 10;
27     int b = 2;
28     int c = a / b;
29     printf("Division Result = %d\n", c);
30
31     int value = 20;
32     int *p = &value;
33     printf("Pointer Value = %d\n", *p);
34     return 0;
35 }
```

Correct Sum
Division Result = 5
Pointer Value = 20
Process exited after 0.1261 seconds with return value 0
Press any key to continue . . .

Compilation results...
Errors: 0
Warnings: 0
Output Filename: C:\Users\google\OneDrive\Desktop\error1.exe
Output Size: 322,614,660 B
Compilation Time: 2.09s

Problem 2:

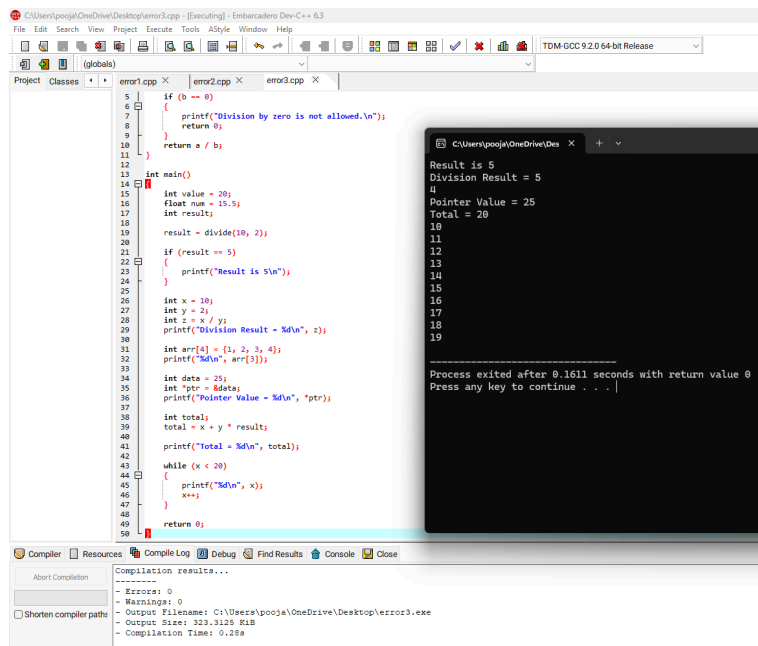
The following C program contains multiple lexical errors introduced intentionally. These errors occur due to invalid tokens, incorrect identifiers, malformed constants, and illegal symbols, which are typically detected during the lexical analysis phase of a compiler. Carefully examine the program and answer the questions given below.



Problem 3:

The following C program is designed to perform arithmetic operations, function calls, conditional checks, pointer usage, and looping. However, the program contains multiple errors distributed across different phases of a compiler, including:

- A. Lexical Errors**
- B. Syntax Errors**
- C. Semantic Errors**
- D. Intermediate Code Generation Issues (logic/precedence)**
- E. Code Optimization Issues (redundancy/inefficiency)**
- F. Runtime / Target Code Errors**



GITHUB:

